ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Fractions in Disguise** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 1200	CP = SP×100/(100+5)
2	**(c)** 1800	CP = SP×100/(100−15)
3	**(c)** 1800	MP = SP×100/(100−15)
4	**(b)** 3000	P = SI×100/(R×T)
5	**(b)** 240	Original = result × 100/125
6	**(b)** 8	Difference = $P(R/100)^2 = 5000\times(4/100)^2$
7	**(a)** 1620	2000×90/100×90/100
8	**(c)** 2000	CP = SP×100/(100+20)
9	**(a)** 3500	CP = SP×100/(100−20)
10	**(d)** 2800	MP = SP×100/(100−20)

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	CP = SP×100/(100+15) (1) = ₹ 1200 (1).	1+1
12	SI = ₹ 840 (1); R = SI×100/(P×T) = 5% (1).	1+1
13	Increase = ₹ 40 (1); % = 40/160×100 = 25% (1).	1+1
14	CP = SP×100/(100+5) (1) = ₹ 2400 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	After first discount: ₹ 3600 (1); SP = ₹ 3420 (1); equivalent discount = 14.5% (1).	1+1+1
16	SI = ₹ 1800 (1); T = SI×100/(P×R) (1) = 3 years (1).	1+1+1
17	After year 1: 10000×110/100 = ₹ 11000 (1); after year 2: ×120/100 = ₹ 13200 (2).	1+2
	OR: After first discount: ₹ 2000 (1); SP = ₹ 1800 (1); equivalent discount = 28% (1).	1+1+1
18	SI = ₹ 1200 (1); T = SI×100/(P×R) (1) = 5 years (1).	1+1+1
	OR: After year 1: 5000×120/100 = ₹ 6000 (1); after year 2: ×110/100 = ₹ 6600 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	CP = 2100×100/105 = ₹ 2000 (2); new SP − CP = ₹ 400 (1); percent = 20% gain (2).	2+1+2
	OR: Total sum = ₹ 10000 (1); R = SI×100/(sum×T) = 5% (2); interest on first = ₹ 400 (1); on second = ₹ 600 (1).	1+2+1+1
20	CP = 3300×100/110 = ₹ 3000 (2); new SP − CP = ₹ -300 (1); percent = -10% loss (2).	2+1+2
	OR: Total sum = ₹ 8000 (1); R = SI×100/(sum×T) = 8% (2); interest on first = ₹ 720 (1); on second = ₹ 1200 (1).	1+2+1+1

Mistakes to avoid:

- Calculating profit or loss percent on the selling price; it is always on the **cost price**.
- Calculating discount on the selling price; it is on the **marked price**.
- Adding two successive percent changes: $+10\%$ then -10% is a 1% decrease, not 0%.
- Using the simple interest formula for compound interest, or forgetting to subtract P from the amount to get CI.
- Applying GST on the marked price when the question says it is charged after the discount.
- Writing a percentage without dividing by 100 in calculations (e.g. 12% of 2000 as 12×2000).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Fractions in Disguise** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 4500	P = SI×100/(R×T)
2	**(d)** 360	Original = result × 100/115
3	**(b)** 20	Difference = P(R/100)² = 8000×(5/100)²
4	**(b)** 3600	5000×80/100×90/100
5	**(b)** 1400	CP = SP×100/(100+25)
6	**(d)** 4000	CP = SP×100/(100−10)
7	**(c)** 1500	MP = SP×100/(100−10)
8	**(b)** 6400	P = SI×100/(R×T)
9	**(a)** 520	Original = result × 100/130
10	**(b)** 9	Difference = P(R/100)² = 10000×(3/100)²

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	SI = ₹ 1176 (1); R = SI×100/(P×T) = 7% (1).	1+1
12	Increase = ₹ 45 (1); % = 45/225×100 = 20% (1).	1+1
13	CP = SP×100/(100+8) (1) = ₹ 4000 (1).	1+1
14	SI = ₹ 3120 (1); R = SI×100/(P×T) = 12% (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	After first discount: ₹ 5100 (1); SP = ₹ 4590 (1); equivalent discount = 23.5% (1).	1+1+1
16	SI = ₹ 2304 (1); T = SI×100/(P×R) (1) = 2 years (1).	1+1+1
17	After year 1: 8000×105/100 = ₹ 8400 (1); after year 2: ×110/100 = ₹ 9240 (2).	1+2
	OR: After first discount: ₹ 2700 (1); SP = ₹ 2160 (1); equivalent discount = 40% (1).	1+1+1
18	SI = ₹ 1400 (1); T = SI×100/(P×R) (1) = 4 years (1).	1+1+1
	OR: After year 1: 20000×110/100 = ₹ 22000 (1); after year 2: ×105/100 = ₹ 23100 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	CP = 4800×100/120 = ₹ 4000 (2); new SP − CP = ₹ 0 (1); percent = 0% loss (2).	2+1+2
	OR: Total sum = ₹ 10000 (1); R = SI×100/(sum×T) = 6% (2); interest on first = ₹ 1680 (1); on second = ₹ 720 (1).	1+2+1+1
20	CP = 2760×100/115 = ₹ 2400 (2); new SP − CP = ₹ 240 (1); percent = 10% gain (2).	2+1+2
	OR: Total sum = ₹ 10000 (1); R = SI×100/(sum×T) = 4% (2); interest on first = ₹ 500 (1); on second = ₹ 1500 (1).	1+2+1+1

Mistakes to avoid:

- Calculating profit or loss percent on the selling price; it is always on the **cost price**.
- Calculating discount on the selling price; it is on the **marked price**.
- Adding two successive percent changes: $+10\%$ then -10% is a 1% decrease, not 0%.
- Using the simple interest formula for compound interest, or forgetting to subtract P from the amount to get CI.
- Applying GST on the marked price when the question says it is charged after the discount.
- Writing a percentage without dividing by 100 in calculations (e.g. 12% of 2000 as 12×2000).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Fractions in Disguise** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 960	1600×75/100×80/100
2	**(b)** 3000	CP = SP×100/(100+10)
3	**(d)** 6000	CP = SP×100/(100−30)
4	**(d)** 2800	MP = SP×100/(100−25)
5	**(b)** 2800	P = SI×100/(R×T)
6	**(c)** 850	Original = result × 100/120
7	**(d)** 40	Difference = $P(R/100)^2 = 4000\times(10/100)^2$
8	**(b)** 2160	3000×90/100×80/100
9	**(c)** 4250	CP = SP×100/(100+8)
10	**(d)** 1000	CP = SP×100/(100−13)

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Increase = ₹ 48 (1); % = 48/320×100 = 15% (1).	1+1
12	CP = SP×100/(100+30) (1) = ₹ 4500 (1).	1+1
13	SI = ₹ 1620 (1); R = SI×100/(P×T) = 15% (1).	1+1
14	Increase = ₹ 30 (1); % = 30/125×100 = 24% (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	After first discount: ₹ 7200 (1); SP = ₹ 5760 (1); equivalent discount = 28% (1).	1+1+1
16	SI = ₹ 3240 (1); T = SI×100/(P×R) (1) = 6 years (1).	1+1+1
17	After year 1: 12500×104/100 = ₹ 13000 (1); after year 2: ×108/100 = ₹ 14040 (2).	1+2
	OR: After first discount: ₹ 1050 (1); SP = ₹ 945 (1); equivalent discount = 37% (1).	1+1+1
18	SI = ₹ 2400 (1); T = SI×100/(P×R) (1) = 8 years (1).	1+1+1
	OR: After year 1: 6000×115/100 = ₹ 6900 (1); after year 2: ×120/100 = ₹ 8280 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	CP = 5850×100/130 = ₹ 4500 (2); new SP − CP = ₹ 450 (1); percent = 10% gain (2).	2+1+2
	OR: Total sum = ₹ 15000 (1); R = SI×100/(sum×T) = 10% (2); interest on first = ₹ 1200 (1); on second = ₹ 1800 (1).	1+2+1+1
20	CP = 4620×100/110 = ₹ 4200 (2); new SP − CP = ₹ 0 (1); percent = 0% loss (2).	2+1+2
	OR: Total sum = ₹ 3000 (1); R = SI×100/(sum×T) = 7% (2); interest on first = ₹ 420 (1); on second = ₹ 630 (1).	1+2+1+1

Mistakes to avoid:

- Calculating profit or loss percent on the selling price; it is always on the **cost price**.
- Calculating discount on the selling price; it is on the **marked price**.
- Adding two successive percent changes: $+10\%$ then -10% is a 1% decrease, not 0%.
- Using the simple interest formula for compound interest, or forgetting to subtract P from the amount to get CI.
- Applying GST on the marked price when the question says it is charged after the discount.
- Writing a percentage without dividing by 100 in calculations (e.g. 12% of 2000 as 12×2000).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.